

WASP-104b

R-2215

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_241218 · created 2026-08-14T23:07:39+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460662.9718 +/- 0.0056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.042
Transit depth (Rp/Rs)^2	2.27 +/- 1.27 [%]
Orbital Inclination (inc)	82.8 +/- 2.23
Airmass coefficient 1 (a1)	1.028 +/- 0.015
Airmass coefficient 2 (a2)	-0.016 +/- 0.042
Scatter in the residuals of the lightcurve fit is	1.5 %
Best Comparison Star	#1 - [606, 83]
Optimal Aperture	3.42
Optimal Annulus	13.26
Transit Duration (day)	0.055 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.042 vs 0.1214 (deviation 0.7σ)
Duration fit vs published	0.055 d vs 0.0742 d (deviation 0.96σ)
Mid-time O–C	1.4 min
Depth SNR	3.6
Residual scatter	1.5 %

Light curve

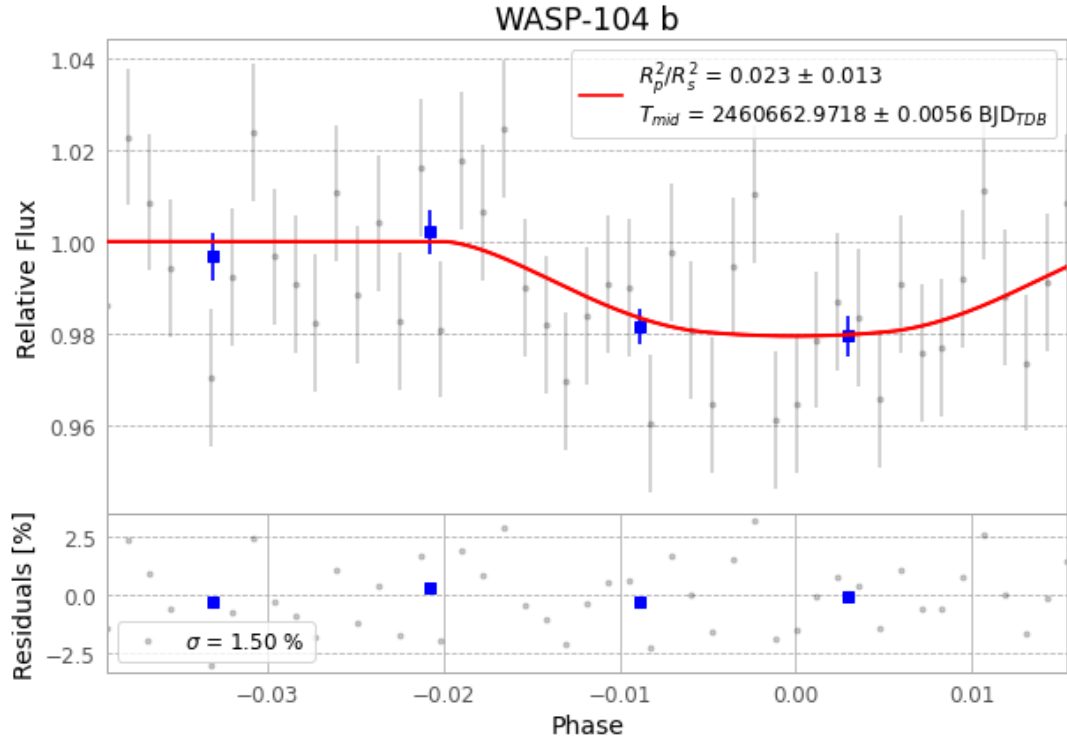


Figure 1 — FinalLightCurve_WASP-104 b_2024-12-18.png (SHA-256 12ba0b00e4f490d0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [401, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 36cb5519a631ea4d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

47 FITS frames (31 MB), WASP-104241218093616.FITS → WASP-104241218115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-241218.log (5d09b79fcfe9...), FinalLightCurve_WASP-104 b_2024-12-18.png (12ba0b00e4f4...), FinalLightCurve_WASP-104 b_2024-12-18.csv (13bc11944592...)

Compute resources

Wall time 123.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 23:07Z.